Differential expansion method for knot invariants

M.Sc. Project Report

Submitted by

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I would like to express my sincere gratitude to my supervisor Prof. Ramadevi for giving an M.Sc. project, guidance, comments, suggestions throughout the project. Thanks to her that she recommended me to Dr. V. Singh, who is former PhD student of her. He has done his PhD in this field and is excellent in it. I have worked on this project after coming home also, both of them have helped me and gave time in this difficult times. He himself said I can call him whenever I have doubt and its very good to have any time approach seniors, he gave me hints to progress and few of steps which were to difficult for me at this level were worked out by him. He has calculated HOMFLY for KT and C knots upto four boxes and important step which is master HOMFLY for three classes in fundamental representation. I was completely stuck when I had to find pattern in HOMFLY for KT knot, both of them helped me, mam gave me hints and V. Singh has suggested many possible ways to find pattern. So thanks to both of them.

Motivation

Knot theory was more like a mathematical theory than physical until Edward Witten's showed it has physical significance. Vacuum expectation value of Wilson loop operator are topological invariants for knots. There is an open problem in knot theory which grabbed my interest and I have decided to do project related to the problem. The problem is distinguishing mutant knots using knot invariants. I have done first stage of MSc project under prof. P. Ramadevi and I came to know she is working on this problem so I thought I should give a try and do this project. My project is basically finding HOMFLY and G functions in symmetric representation for 10 and 11 crossing knots and including twist knots in it so that one has master HOMFLY with some parameters in it. Different parameter correspond to different knot. If I find some result, I may also tell relation between G functions of three class. There is a related problem, which is, to generalize this result to rectangular and nonrectangular representation, which I will do in future. This result has its use in knot-quiver correspondence and string theory. Knot theory has dramatic history and beautiful properties. Knot theory earlier thought as a theory of unifying particles for example different knots were thought as different particles in ether medium. Later its proved that there is nothing called as ether medium. Knot theory is closely related to string theory in which I have a huge interest as its the theory of unification. After knowing interesting things about this theory I have decided to do this project.

Contents

Chapter 1

Introduction

In knot theory, knot invariant is a quantity defined for each knot and it is same for equivalent knots. Knot invariant could be number or polynomial. There is a widely use knot invariant called HOMFLY-PT, its basically Laurent polynomial. Now computing this knot invariant is very useful and interesting as there will be unique identity for each knots. HOMFLY can be computed in different representation(R), for R equals single box its called uncoloured HOMFLY and vice-versa. There are various ways to compute HOMFLY but I will review differential expansion method(DE). First by using differential expansion method I will obtain coloured HOMFLY for class of knots called twist knots. I plan to extend this methodology to other classes of knots which are 10 and 11 crossing knots.

In differential expansion method itself there are different tricks to find HOMFLY. Chapter 2 is basically one such trick played to find coloured HOMFLY for a twist knot. I try to extend this trick to a 11 crossing knot. Chapter three is unrelated to aim of the project but it is important for broader result which is finding HOMFLY in non-rectangular representation. As I said there are different tricks to find coloured HOMFLY, in chapter four I will present second trick applied on twist knots and try to extend it to 10 and 11 crossing knots.

Chapter 2

Review of differential expansion for figure eight knot and HOMFLY for defect -1 knots

One can classify knots on the basis of defects. Defect for particular knot is the highest power of q^2 in Alexander polynomial for that knot. Alexander polynomial is equal to HOMFLY at $A = 1$, A is a variable of HOMFLY. One may calculate HOMFLY in different representation R and for $R > [1]$ its called coloured HOMFLY and vice-versa. One box is represented by [1], two boxes by [2] and so on, these are called symmetric representations($[r]$). In defect zero knot class there are two famous knots figure eight(4_1) and trefoil(3_1), which I will mostly use. In this chapter, I will tell how differential expansion is applied to trefoil to find coloured HOMFLY and then I will try to extend it to defect -1, 11 crossing knots. I will first try to find HOMFLY in general symmetric representation for Kinoshita Tarasake and Convey knot(KT and C), which is defect -1, 11 crossing mutant knot pair and try to find HOMFLY for other 11 crossing knots. Knots are said to be mutant if their knot invariant in symmetric representation coincide. Further in this chapter, I will tell restrictions on differential expansions from group theory in which one will see how two representations in a particular SU(N) are related. The essence of this mathematics is to derive coloured HOMFLY(2.3) for any knot in terms of unknown G functions. In the following sections I will tell how to generalize HOMFLY and use it to generalize coloured HOMFLY for KT and C knot.

2.1 Restriction on differential expansion from group theory

We will use three properties

For $A = q^N, H_R(A, q)$ depends on representation R of $SU(N)$

For $A = q^{-N}$, the transposed $H_R(A, q^(-1)) = H_{tr}(A, q)$ depends on representation R^{tr} of $SU(N)$, where R^{tr} is transpose of R.

For particular $SU(N)$ there are special relations between conjugate representations with l and $N - l$ lines.

For rectangular young diagram $[r^s]$ representation $[r^s] \cong \emptyset$ for $SU(s)$ and similarly $\overline{[r^s]} = [s^r] \cong \emptyset$, where overline is used for transposition of young diagram. Then the above three facts imply that

$$H_{[r^s]}(A = q^s, q) = 1 \quad and \quad H_{[r^s](A=q^{-r},q)} = 1 \tag{2.1}$$

More generally

$$[r^s] \cong [r^{N-s}] \quad and \quad \overline{[r^s]} = [s^r] \cong [s^{N-r}] = \overline{[(N-r)^s]} \tag{2.2}$$

what imposes severe constraints on differential expansion(DE). Differential is basically expanding HOMFLY(H) into $differentials$ $D_n = \{Aq^n\}$ or $quadratic$ $differentials$ $Z^{(i)}_{[r]} = \{Aq^{r+i}\}\{Aq^{i-1}\}$, where $\{x\} = x - x^{-1}$. Now by using all the above properties one gets [4]

$$H^{\kappa}_{[r]} = \sum_{k=0}^{r} \frac{[r]!}{[k]![r-k]!} G^{\kappa}_k(A,q) \left(\prod_{i=0}^{k-1} \{Aq^{r+i}\} \right) \{A/q\} \tag{2.3}$$

with $G^{\kappa}_0 = 1/\{A/q\}$ and κ represents different knots. This is the generic form of symmetric differential expansion. The transposed version for antisymmetric representation($q \to q^{-1}$) is

$$H^{\kappa}_{[1^r]} = \sum_{k=0}^{r} \frac{[r]!}{[k]![r-k]!} G^{\kappa}_k(A,q^{-1}) \left(\prod_{i=0}^{k-1} \{A/q^{r+i}\} \right) \{Aq\} \tag{2.4}$$

with $G^{\kappa}_0 = 1/\{Aq\}$. The coefficients G_k are further factorized to

$$G^{\kappa(d)}_k(A,q) = F^{\kappa(d)}_k(A,q) \prod_{i=1}^{k-d-1} \{Aq^{i-1}\} \tag{2.5}$$

the parameter d is the defect of the DE and it was conjectured that it is equal to degree of alexander polynomial minus one. In the defect zero case H is simplified if written in quadratic differential $Z^(i)_[r]$ instead of differential D_n i.e.

$$H^{\kappa(0)}_{[r]}(A,q) = \sum_{k=0}^{r} \frac{[r]!}{[k]![r-k]!} F^{\kappa(0)}_{[r]}(A,q) \prod_{i=0}^{k-1} Z^{(i)}_{[r]}. \tag{2.6}$$

2.2 Group theory restriction for rectangular diagrams

In this section first we will see few properties and write down skeleton of H in terms of quadratic differential, quantum numbers $[n] = (q^n - q^{-n})/(q - q^{-1})$ and some question marks. Then by comparing this H with already known results one find full form of H. Now by using (2.2) one may write for $R = [33] = [3^2]$ as

$$H_{[33]} - \underbrace{1}_{degree\ 0} \sim \{A/q^2\} \Longleftarrow [33] \cong \emptyset \quad SU(2) \tag{2.7}$$

$$H_{[33]} - \underbrace{1}_{degree\ 0} \sim \{Aq^3\} \Longleftarrow [222] = \overline{[33]} \cong \emptyset \quad SU(3) \tag{2.8}$$

$$H_{[33]} - \underbrace{H_{[11]}} \sim \{Aq^4\} \Longleftarrow [222] = \overline{[33]} \cong [2] = \overline{[11]} \quad SU(4) \tag{2.9}$$

$$H_{[33]} - \underbrace{H_{[3]}}_{degree\ 3} \sim \{A/q^3\} \Longleftarrow [33] \cong [3] \ SU(3) \tag{2.10}$$

$$H_{[33]} - \underbrace{H_{[22]}}_{degree\ 4} \sim \{Aq^5\} \Longleftarrow [222] = \overline{[33]} \cong [22] = \overline{[22]} \quad SU(3) \tag{2.11}$$

which implies that

$$H^{\kappa}_{[33]} = 1 + \{Aq^3\}\{A/q^2\}([3][2]G_1 + ?\{?\} + \{Aq^4\}(?\{?\} + \{A/q^3\}(\{?\} + \{Aq^5\}(\{?\} + ?\{?\}\{?\})))).$$

All the differentials from above five equations are arranged in H according to their degrees, unknown differentials and coefficients are question marks. Now converting the above H into quadratic and F function as

$$H^{\kappa(0)}_{[33]} = 1 + Z^{(0)}_{[33]}([3][2]F_1 + ?Z^{(?)} + Z^{(1)}_{[33]}(?Z^{(?)}_{[33]} + Z^{(-1)}_{[33]}(?Z^{(?)}_{[33]} + Z^{(2)}_{[33]}(?Z^{(?)}_{[33]} + ?Z^{(?)}_{[33]}Z^{(?)}_{[33]})))).$$

Still the coefficients and powers are unknown which one can find by comparing it with already known results. For example we already know result for trefoil(3_1) knot[4] and after comparing it with above H we get

$$H^{3_1}_{[33]} = 1 - [3][2]A^2Z^{(0)}_{[33]} + ([3]^2q^2A^4Z^{(+1)}_{[33]} + \frac{[3][4]}{[2]}q^{-2}A^4Z^{(-1)}_{[33]})Z^{(0)}_{[33]} - ([4]q^6A^6Z^{(+2)}_{[33]} + [4][2]^2A^6Z^{(-1)}_{[33]})Z^{(+1)}_{[33]}Z^{(0)}_{[33]} +$$

$$([3]^2q^4A^8Z^(+ 2)_{[33]} + \frac{[3][4]}{[2]}A^8Z^{(0)}_{[33]})Z^{(+1)}_{[33]}Z^{(0)}_{[33]}Z^{(+1)}_{[33]} - [3][2]q^4A^{10}Z^{(+2)}_{[33]}Z^{(+1)}_{[33]}(Z^{(0)}_{[33]})^2Z^{(-1)}_{[33]} +$$

$$q^6A^{12}Z^{(+2)}_{[33]}(Z^{(+1)}_{[33]}Z^{(0)}_{[33]})^2Z^{(-1)}_{[33]}.$$

In this way one can find for HOMFLY-PT in [33], [44], so on until one generalizes HOMFLY in general $[rr]$. Similarly by finding H in [333], [444], and so on until one generalizes to $[rrr]$ representation. Similarly one finds H in general $[r^n]$ representation[4]. In this chapter, one clearly sees the technique to generalize HOMFLY to any symmetric representation $[r]$ by finding pattern in [1], [2], [3], ... and eventually generalizing to $[r]$, its like induction.

2.3 An Example of knot invariant for defect -1 and an attempt to generalized it for such knots

Now I have used the induction technique to find HOMFLY for KT and C knot in symmetric representation. KT and C are mutant knots under 11 crossing class. We thought we will find HOMFLY for KT and C knot in general symmetric representation $[r]$ and by further generalizing we will include other 11 crossing knots. So that in the end we have HOMFLY for 11 crossing knots in symmetric representations. My guide and Dr. V. Singh provided me HOMFLY in four boxes $[1],\ [2],\ [3],\ and\ [4]$, here are first two and rest two I have not shown because they are lengthy and will take two pages of my report
$H_{[1]} = \frac{q^4}{A^4} + \frac{1}{A^4q^4} - \frac{q^2}{A^4} - \frac{1}{A^4q^2} + \frac{2}{A^4} - \frac{q^6}{A^2} - \frac{1}{A^2q^6} - A^2q^4 - \frac{A^2}{q^4} + A^2q^2 - \frac{2q^2}{A^2} + \frac{A^2}{q^2} - \frac{2}{A^2q^2} - 2A^2 + q^6 + \frac{1}{q^6} + 2q^2 + \frac{2}{q^2} + 1$

$H_{[2]} = \frac{1}{A^8q^{20}} - \frac{1}{A^8q^{18}} - \frac{1}{A^8q^{16}} + \frac{3}{A^8q^{14}} + \frac{1}{A^8q^{12}} - \frac{3}{A^8q^{10}} + \frac{2}{A^8q^8} + \frac{2}{A^8q^6} + \frac{q^4}{A^8} - \frac{2}{A^8q^4} - \frac{q^2}{A^8} + \frac{1}{A^8q^2} + \frac{1}{A^8} - \frac{1}{A^6q^{22}} + \frac{1}{A^6q^{18}} - \frac{3}{A^6q^{16}} - \frac{2}{A^6q^{14}} - \frac{1}{A^6q^{12}} - \frac{q^{10}}{A^6} - \frac{2}{A^6q^{10}} - \frac{q^8}{A^6} + \frac{1}{A^6q^8} - \frac{6}{A^6q^6} - \frac{q^4}{A^6} - \frac{4}{A^6q^4} - \frac{4q^2}{A^6} + \frac{3}{A^6q^2} - \frac{3}{A^6} + \frac{1}{A^4q^{22}} + A^4q^{18} + \frac{1}{A^4q^{18}} - A^4q^{16} + \frac{2}{A^4q^{16}} - A^4q^{14} + \frac{2q^{14}}{A^4} + \frac{4}{A^4q^{14}} + 3A^4q^{12} + \frac{3}{A^4q^{12}} + \frac{q^{10}}{A^4} - 3A^4q^8 + \frac{5q^8}{A^4} + \frac{7}{A^4q^8} + 3A^4q^6 + \frac{6q^6}{A^4} + \frac{A^4}{q^6} + \frac{13}{A^4q^6} + A^4q^4 - \frac{A^4}{q^4} - \frac{3}{A^4q^4} - 2A^4q^2 + \frac{4q^2}{A^4} + \frac{A^4}{q^2} + \frac{3}{A^4q^2} + 2A^4 + \frac{15}{A^4} - A^2q^{20} - \frac{1}{A^2q^{20}} - \frac{q^{18}}{A^2} - \frac{1}{A^2q^{18}} + A^2q^{16} - \frac{q^{16}}{A^2} - 3A^2q^{14} - \frac{5}{A^2q^{14}} - A^2q^{12} - \frac{5q^{12}}{A^2} - \frac{A^2}{q^{12}} - \frac{3}{A^2q^{12}} - \frac{5q^{10}}{A^2} - \frac{A^2}{q^{10}} - 5A^2q^8 + \frac{q^8}{A^2} - \frac{15}{A^2q^8} + A^2q^6 - \frac{12q^6}{A^2} - \frac{2A^2}{q^6} - \frac{5}{A^2q^6} - 2A^2q^4 - \frac{14q^4}{A^2} - \frac{6A^2}{q^4} + \frac{4}{A^2q^4} - 9A^2q^2 + \frac{5q^2}{A^2} - \frac{22}{A^2q^2} + A^2 - \frac{12}{A^2} + q^{20} + q^{16} + \frac{2}{q^{16}} + 2q^{14} + 3q^{12} + \frac{1}{q^{12}} + 5q^{10} + \frac{6}{q^{10}} + 3q^8 + \frac{5}{q^8} + 2q^6 - \frac{3}{q^6} + 15q^4 + \frac{12}{q^4} + 7q^2 + \frac{14}{q^2} - 3$

As I said, I had to find pattern in these four HOMFLY and generalize it to $[r]$. I tried varies things in mathematica like Simplify, Together, Collect, Expand, and so on functions. Then I have tried finding pattern in $G_1,\ G_2,\ G3,\ and\ G_4$ functions because if we generalize G also we can find HOMFLY in symmetric representation. For finding pattern, I have also used website [1] for integer sequence. I couldn't find pattern by any of these things and then decided to find HOMFLY for symmetric representation from different technique, which is mentioned in chapter four. In this chapter I have reach dead-end!

Chapter 3

HOMFLY in non-rectangular representation

There two more representations other than symmetric which are rectangular and non-rectangular representation. Rectangular representations are like $[rr]$, $[rrr]$, $[r^4]$, so on and non-rectangular representation are like $[rs]$ for $r \neq s$. In this project I am trying to find HOMFLY for 10 and 11 crossing knots in symmetric representation. Future work and broader aim is to find HOMFLY for these knots in rectangular and non-rectangular representations. In this chapter I will review HOMFLY in non-rectangular representation so that one gets insight of it, this chapter can be skipped. HOMFLY for double braid in non-rectangular representation can be written as [5]

$$H_R^{(m,n)} = \sum_{X \subset R \otimes \overline{R}} Z_R^X \mathcal{F}_X^{(m,n)} = \sum_{X \subset R \otimes \overline{R}} Z_R^X \cdot F_X^{(m)} F_X^{(n)} \tag{3.1}$$

where Z depends only on representation and $\mathcal{F}$ depends on representation as well as on knot. $F_X^{(m)}$ can be written as [3]

$$F_X^{(m)} = \sum_{Y \triangleleft X} (\mathcal{B}^{m+1})_{XY} \tag{3.2}$$

where $\mathcal{B}$ is a universal triangular "embedding" matrix with $Y \triangleleft X \subset R \otimes \overline{R}$. Above two equations are actually defined for rectangular representations but in [7] they have extended it to non-rectangular representation with increased elements in $\mathcal{B}$ matrix. Representation X and Y are composite . For rectangular representations $R = [r^n]$ only very special diagonal composites contribute to $R \otimes \overline{R}$ and embedding for diagonal composites is understood as embedding of the corresponding λ:

$$(\mu, \mu) \triangleleft (\lambda, \lambda) \Longleftrightarrow \mu \subset \lambda. \tag{3.3}$$

The entries of the matrix $\mathcal{B}$ in (3.2) are expressed through skew schur functions [7]

$$\mathcal{B}_{\lambda\mu} = (-1)^{|\lambda|-|\mu|} \cdot \Lambda_\lambda \cdot \frac{\chi_\mu^0 \cdot \chi_{\lambda^\vee/\mu^\vee}^0}{\chi_\lambda^0} \tag{3.4}$$

where $\vee$ stands for transposition of the young diagram and Λ_μ are the eigenvalues of the $\mathcal{R}$ matrix in the channel

$$R \otimes \overline{R} = \oplus_{\mu \in R} (\mu, \mu) \tag{3.5}$$

best express through hook parameters of $\lambda = (a_1, b_1 | a_2, b_2 |, ...)$:

$$\Lambda_\lambda = \prod_i^{total\ \ hooks(\lambda)} (q^{a_i - b_i A} A)^{2(a_i+b_i+1)} \tag{3.6}$$

We will consider the simplest case of $[r,1]$ for which answer is already known from [6]. In addition to $2r+1$ diagrams $X=(\lambda,\ \lambda)$ with $\lambda\subset R=[r,1]$ i.e. $\lambda=\emptyset,\ [i],\ [i,1],\ i=1,...r$ there are $r-1$ additional composite pairs $\tilde{X}_i=([i-1,1],[i])\oplus([i],[i-1,1])$ with same eigenvalue which wasn't there for rectangular diagram. All the other elements of $\mathcal{B}$ can be found from (3.4) and the property that elements of $\mathcal{B}_{\lambda\mu}$ vanishes if μ is not a sub-diagram of λ, except the followings

$$\mathcal{B}_{\tilde{X}_i,[i]}=(-1)^i\tilde{\Lambda}_i\cdot\frac{[i+1\cdot A^2]}{q^{(i-2)}} \tag{3.7}$$

$$\mathcal{B}_{\tilde{X}_i,[1,1]}=(-1)^i\tilde{\Lambda}_i\cdot\frac{A^2-q^6}{q^{i^2-3i+4}(q^4-1)} \tag{3.8}$$

$$\mathcal{B}_{\tilde{X}_i,[i]}=-\tilde{\Lambda}_i\cdot\frac{A^2q^{4i-2}-1}{q^{2i}-1} \tag{3.9}$$

$$\mathcal{B}_{\tilde{X}_i,[i,1]}=0 \tag{3.10}$$

In the simplest case of R=[2, 1] the matrix is

$$\mathcal{B}^{[2,\ 1]}=\begin{array}{c|cccccc} & \emptyset & [1] & [1,1] & [2] & [2,1] & \tilde{X}_2\\ \hline \emptyset & 1 & 0 & 0 & 0 & 0 & 0\\ [1] & -A^2 & A^2 & 0 & 0 & 0 & 0\\ [1,1] & \frac{A^4}{q^2} & -\frac{[2]A^4}{q^3} & \frac{A^4}{q^4} & 0 & 0 & 0\\ [2] & q^2A^4 & -[2]q^3A^4 & 0 & q^4A^4 & 0 & 0\\ [2,1] & -A^6 & [3]A^6 & -\frac{[3]A^6}{[2]q} & -\frac{[3]qA^6}{[2]} & A^6 & 0\\ \tilde{X}_2 & -A^6 & [3]A^6 & \frac{(A^2-q^6)A^4}{q^2(q^4-1)} & -\frac{(A62q^6-1)A^4}{q^4-1} & 0 & A^4 \end{array} \tag{3.11}$$

The first row and first column are for reference, rest is actual matrix for $[2,1]$. The new one added for the non-rectangular case is last line in the matrix. Similarly one can find $\mathcal{B}$ matrix for $R=[3,1]$ using (3.4), (3.7)-(3.10). Now that we know $\mathcal{B}$ matrix, from (3.1) and (3.2) we can finally find HOMFLY if we know Z_R^X. Z_R^X for $R=[r,1]$ is already known [6]

$$Z_{[r,1]}^{[k]}=\frac{[r+1]!}{[r][k]![r-k]!}\cdot\frac{D_{r+k-2}!D_{k-2}!}{D_{r-1}}(D_{r-2}D_{k-1}-[r+1][k](q-q^{-1})^2) \tag{3.12}$$

$$Z_{[r,1]}^{[k,1]}=\frac{[k][r+1]!}{[r][k+1]![r-k]!}\cdot\frac{D_{r+k-1}!D_{k-1}!}{D_{r-1}!D_{k-2}!}D_{-1}D_{-2}D_{-3} \tag{3.13}$$

$$Z_{[r,1]}^{\tilde{X}_k}=-(q-q^{-}1)^4\cdot\frac{[r+1]^2[r-1]!}{[r-k]![k-2]!}\cdot\frac{D_{r+k-2}!D_{k-3}!}{D_{r-1}!}\cdot D_{-2} \tag{3.14}$$

Now from (3.1) and (3.2) one can find HOMFLY for $R=[r,1]$ representation and it is yet to be generalized to $R=[r,s]$. As I said this is not related to aim of this project but its related to broader result which is to find HOMFLY for 10 and 11 crossing knots in non-rectangular representation. One of the possible way to generalize present result for twist knot to other knots is to define more parameters in (3.1) and find corresponding Z and F functions, which I will do in future.

Chapter 4

Finding higher G functions for twist knot, 10 and 11 crossing knots

From (2.3) one can see HOMFLY is esily computed if G functions are known. In this chapter, first I will tell how to find higher G functions G_2, G_3, so on using G_1, bilinear operator and multi-linear operator. Bilinear operator will tell how two G_1 are combined to form G_2 and multi-linear operator will tell how more than three G_1 are combined to form hogher G's. These bilinear and multi-linear operator are beautifully defined for twist knots in [2]. It is this paper which gave me idea to define operators for other knots and hence one will be able to find higher G functions for other knot and hence coloured HOMFLY. All $G_j^{(k)}|_{q=1}$ is just power of $G_1^{(k)}|_{q=1}$.

$$G_j^{(k)}|_{q=1} = (G_1^{(k)}|_{q=1})^j \tag{4.1}$$

where k's are representing different twist knots for trefoil $k = 1$, for figure eight $k = -1$ so on. Note this relation is true at $q = 1$, to generalize it to all q, one need to define operator between G_1 functions.

$$G_j^k = \circ(G_1^k)^j \tag{4.2}$$

and for $j = 2$ it is

$$G_2^k = G_1^k \circ G_1^k \tag{4.3}$$

such that at $q = 1$ this equation is back to (4.1). Now there are two different operators one is bilinear operator which is between two G_1 function and other is multi-linear operator, it is between more than two G_1. We will need one important equation for following arguments, which is G_1 function for twist knot [2]

$$G_1^{(k)} = A\frac{1 - A^{2k}}{\{A\}} \tag{4.4}$$

4.1 Bilinear operator($\circ_1$) for twist knot

Note from (4.2) and (4.3), defining operator between G function is basically defining operator between A's. Its given as

$$for\ m \leqslant n \qquad A^{2m} \circ_1 A^{2n} = q^{4m-2}A^{2m+2n} \tag{4.5}$$

$$for\ m > n \qquad A^{2m} \circ_1 A^{2n} = q^{4n}A^{2m+2n} \tag{4.6}$$

. When two G_1 functions(4.3) are combined with this operator we get following G_2 function

$$G_2^{(k)} = qA^2\left(\frac{1}{\{A\}\{Aq\}} - [2]\frac{A^{2k}}{\{A\}\{Aq^2\}} + \frac{q^{4k}A^{4k}}{\{Aq\}\{Aq^2\}}\right) \tag{4.7}$$

4.2 Multi-linear operator($\circ$) for twist knot

Multi-linear case is when more than two G functions or A's combine together. When one tries to define multi-linear operator for A's, they will get some coefficients and powers of A's. Powers of A's are conserved in both operators, whichever powers of A's are combining one gets the same powers of A back

$$\circ(A^{k_1},\ A^{k_2}, ...\ A^{k_n}) = C_k A^{\sum_{i=1}^{n} k_i} \tag{4.8}$$

C_k further breaks to

$$C_k = c_k^a c_k^f \tag{4.9}$$

where c_k^a is average power of q which depends on k but not on permutations and c_k^f is called fine structure which depends on permutation with repetitions. c_k^f is defined as

$$c_k^f = q^{2(total\ \ inversions) - max(total\ \ inversions)} \tag{4.10}$$

where $total\ inversions$ counts inversions in the permutation with repetition and with different arrangement of $\{k_1, k_2, ..., k_n\}$. There will be one arrangement of $k's$ such that $total\ inversions$ is maximum and its called $max(total\ inversions)$. Now for average, let D be the young diagram with boxes $\sum_{i=1}^{n} k_i$. Arranging $\{k_1, k_2, ..., k_n\}$ set as: $k_1 \geqslant k_2 \geqslant k_3 \geqslant .... \geqslant k_n$ such that k_i is the i^{th} box of the young diagram D. The above set can be parametrized to $\{k_{i_1}^{m_1},\ k_{i_2}^{m_2},\ ....,\ k_{i_l}^{m_l}$, here m_j is the multiplicity of k_{i_j} with $\sum_{j=1}^{l} m_j = n$. Make the young diagram of multiplicity m_i, its called d and these $m_i's$ make individual floor of d in decreasing order. The claim is

$$c_k^a = q^{2\nu(D) - n(n-1)/2 + \nu(d)} \qquad when \quad \forall i,\ k_i > 0 \tag{4.11}$$

$$c_k^a = q^{2\nu(D) - n(n-1)/2 - \nu(d)} \qquad when \quad \forall i,\ k_i < 0 \tag{4.12}$$

with $\nu(D) = \sum_{j=1}^{l} (j-1) D_i,\ D_i$ is individual floors of young diagram D.

Now one knows the operators in (4.2) which are bilinear and multi-linear operator, one can find higher G functions for twist only from G_1. (4.2) and (4.4) shows how beautifully whole twist family is combined to each other. I am trying to define these bilinear and multi-linear operator for 10 and 11 crossing knots and trying to establish relation between three class of knots twist knots, 10 and 11 crossing knots. In the following section I will tell the procedure to find these operators.

4.3 Procedure for calculating higher G functions

I will tell the steps which I will follow to find generalized bilinear and multi-linear operator for three classes in chronological order. So far I have succeeded step 1 and on step 2 and I will continue to follow these steps even after report also until I reach dead-end. I think most likely I will find G_2 function. Higher G functions which involve multi-linear operator looks highly non-trivial at the moments and requires little more thinking.

1. **Calculating HOMFLY in fundamental representation:**

   First step is to calculate master HOMFLY in fundamental representation which includes all three classes twist knot, 10 and 11 crossing knots. Dr. V. Singh has already calculated HOMFLY in fundamental representation and it has multiple parameters in it such that each parameter corresponds to some knot in this classes. He has calculated through Wilson loop average in S_3 manifold.

2. **Generalizing this HOMFLY upto four boxes:**

   By generalizing HOMFLY upto four boxes, one can directly get $G_1,\ G_2,\ G_3,\ and\ G_4$ from differential expansion. As I said I am going to find higher G function using (4.2) i.e.

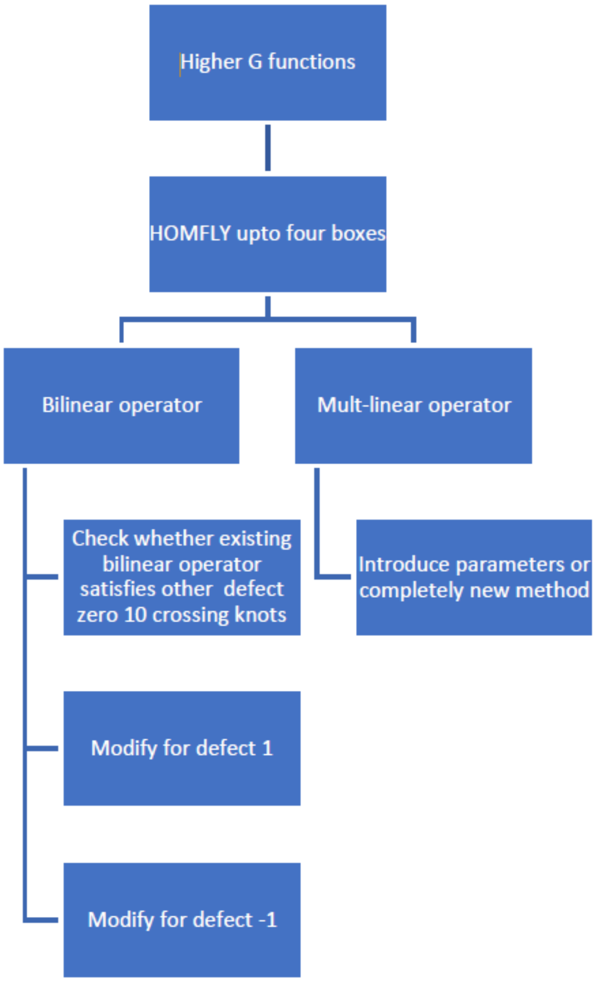
only using G_1 function, to check whether my higher G functions are correct through this relation I am going to cross verify it with G functions which are computed with differential expansion. Moreover computing upto G_4 will help me guessing the form of operators.

3. **Guessing bilinear operator:**

   This step itself has three steps in it first, we already know bilinear operator for twist knot which is (4.5) and (4.6) and twist knot has defect zero, so I think it may be possible that I don't have to modify this bilinear operator for defect zero 10 crossing knots. I will check this and modify it if necessary for defect zero 10 crossing knots. Second, I will try to modify this operator for defect 1 knots by introducing this defect in operator. As putting parameter and generalizing is very strong idea and its used step to find HOMFLY in fundamental representation. In third step I will modify it for defect -1. In this way I will construct bilinear operator.

4. **Guessing multi-linear operator:**

   One of the possible way to generalize multi-linear operator (4.8) to include three classes is similar to bilinear operator by introducing defect parameter in it. If its not in this way then I will try to develop completely new method to find multi-linear operator for three classes of knots.



Chapter 5

Conclusion

I have used two different tricks to find coloured HOMFLY, first is presented in chapter two. I basically showed how to find patterns in the coefficients of HOMFLY, which is induction method to find HOMFLY. Then I try to apply this trick to a 11 crossing mutant pair called Kinoshita-Tarasake(KT) and Convay(C) knot and reached dead-end. Then I decided to use another trick to find coloured HOMFLY which I showed in fourth chapter. This is a great method as it only uses G_1 function, bilinear and multi-linear operator to find all other G functions in (2.3) and hence coloured HOMFLY. I have laid down all the steps which I am going to follow in section (4.3). First step, which is HOMFLY upto four boxes for 10 and 11 crossing knots is very difficult for me at this stage and therefore I am dependent on my guide and V. Singh for this result. As soon as I get this I will start searching for bilinear and multi-linear operator and obtain G functions and HOMFLY for 10 and 11 crossing knots. As soon as I find HOMFLY in symmetric representation I will try to find it in others two representations also which are rectangular and non-rectangular as I said in chapter three. As far as uses of these results are concern it can be used to differentiate mutant knots, which is open problem in knot theory. It could be used in knot-quiver correspondence and string theory as well.

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